

```
1  #include <stdio.h>
2  int main()
3  {
4      int numberOfProcesses;
5      printf("Enter number of processes: ");
6      scanf("%d", &numberOfProcesses);
7      int timeQuantum;
8      printf("Enter time quantum: ");
9      scanf("%d", &timeQuantum);
10
11     int burstTime[10];
12     int remainingTime[10];
13     int waitingTime[10] = {0};
14     int turnaroundTime[10];
15
16     // Read burst times
17     for(int i = 0; i < numberOfProcesses; i++)
18     {
19         printf("Enter burst time of process P%d: ", i + 1);
20         scanf("%d", &burstTime[i]);
21         remainingTime[i] = burstTime[i];
22     }
23
24     int currentTime = 0;
25     int allDone = 0;
26
27     // Round Robin execution using WHILE
28     while(!allDone)
29     {
30         allDone = 1;
31         for(int i = 0; i < numberOfProcesses; i++)
32         {
33             if(remainingTime[i] > 0)
34             {
35                 allDone = 0;
36                 if(remainingTime[i] > timeQuantum)
37                 {
38                     currentTime += timeQuantum;
39                     remainingTime[i] -= timeQuantum;
40                 }
41                 else
42                 {
43                     currentTime += remainingTime[i];
44                     waitingTime[i] = currentTime - burstTime[i];
45                     remainingTime[i] = 0;
46                 }
47             }
48         }
49     }
50
51     int totalWaitingTime = 0;
52     int totalTurnaroundTime = 0;
53
54     // Calculate turnaround time and totals
55     for(int i = 0; i < numberOfProcesses; i++) {
56         turnaroundTime[i] = waitingTime[i] + burstTime[i];
57         totalWaitingTime += waitingTime[i];
58         totalTurnaroundTime += turnaroundTime[i];
59     }
60
61     // Output
62     printf("\nPID\tBT\tWT\tTAT\n");
63     for(int i = 0; i < numberOfProcesses; i++) {
64         printf("P%d\t%d\t%d\t%d\n",
65             i + 1,
66             burstTime[i],
67             waitingTime[i],
68             turnaroundTime[i]);
69     }
70
71     printf("Average Waiting Time = %.2f\n",
72         (float)totalWaitingTime / numberOfProcesses);
73     printf("Average Turnaround Time = %.2f\n",
74         (float)totalTurnaroundTime / numberOfProcesses);
75 }
```